

Nowhere Analyticity of Every Positive Compositional Iterate of the Inverse Fabius Function

Formal reversion, analytic inversion, and the dyadic-spine theorem

Research report prepared for the ProveIt Fabius-function corpus

30 August 2026

Result status. The inverse nowhere-analytic conclusion is already a corollary of the corrected forward-iterate report; this derived companion supplies the expanded formal-radius, center-jet, and iterated-endpoint analysis. A hostile repository review ported the forward report’s uniform defect estimate, corrected two-spine hypothesis, literal $n = 1$ empty tie set, and live module name. The one-fold inverse analytic-locus and endpoint Hölder obstructions have exact Lean counterparts. The finite positive-list defect arithmetic is also exhaustively formalized in `PartitionDefect.lean`, but the set-partition wrapper and weighted Bell/spine estimates are not. The forward and inverse iterate theorems for $n \geq 2$, formal-radius transport, iterated endpoint scale, and all report conjectures remain manuscript mathematics. The numerical script reproduced all seven outputs byte-for-byte in a pinned environment; this finite-order replay is not proof. Former Conjecture 14.2 is already discharged, and former Conjecture 14.4 duplicates the forward report. The inverse clause of former Conjecture 14.1 is false because formal reversion does not preserve eventual jet vanishing. All three appear below only as numbered quarantine warnings.

Abstract

Let $F: [0, 1] \rightarrow [0, 1]$ be the bounded Fabius function, let $I = F^{-1}$, and write

$$G_n = F^{\circ n}, \quad K_n = I^{\circ n} = G_n^{-1}.$$

We prove that for every integer $n \geq 1$, the inverse iterate K_n is real analytic at no point of $[0, 1]$. The statement is strengthened in two directions. First, there is a co-countable dense set $Y_n \subset (0, 1)$ on which the formal Taylor series of K_n has radius of convergence zero. Second, at either endpoint K_n is not even Hölder continuous of any positive order. At the exceptional fixed point $1/2$, by contrast, the complete Taylor series is the affine polynomial

$$\frac{1}{2} + 2^{-n} \left(y - \frac{1}{2} \right),$$

but this polynomial does not represent K_n on any neighborhood. Thus the proof carefully separates nowhere analyticity from divergence of the formal Taylor series.

The interior argument rests on a complete proof of the forward-iterate theorem. The exact derivative atlas

$$F^{(m)}(x) = Q_m \Theta(2^m x), \quad Q_m = 2^{\binom{m+1}{2}},$$

is inserted into the set-partition form of Faà di Bruno’s formula. A quadratic partition defect suppresses every genuinely branched chain-rule tree, leaving a finite sum of orbit spines. Strictly monotone orbit slopes produce a unique dominant spine away from a finite tie set, and binary digit transitions keep its Thue–Morse amplitude bounded away from zero along infinitely many orders. Hence G_n has zero Taylor radius on a co-countable dense set E_n . A general convergence-invariance theorem for formal series under smooth local inversion then gives $Y_n = G_n(E_n)$ and the corresponding zero-radius conclusion for K_n .

The report also derives the iterated endpoint scale

$$-\log K_n(y) \sim (2 \log 2)^{1-2^{-n}} (\log(1/y))^{2^{-n}}, \quad y \downarrow 0,$$

explains the q -binomial, sinc-product, Bell–Bernoulli, Lambert– W , and Legendre connections, supplies reproducible arbitrary-precision series-reversion experiments, and records concrete conjectures and a Lean formalization plan.

Contents

1	Statement of the problem and strengthened results	4
2	Repository audit and novelty boundary	5
2.1	Canonical sources inspected	5
2.2	Existing results and the novelty boundary	5
3	Fabius, Rvachev, Thue–Morse, and the derivative atlas	6
3.1	The Rvachev bump and the signed global extension	6
3.2	Exact all-order derivatives	6
3.3	Shape and orbit dynamics	7
4	Why a direct dyadic-jet induction fails	7
5	The dyadic defect of a Faà di Bruno partition	8
6	A two-spine composition lemma	10
7	The orbit-spine asymptotic for compositional iterates	11
8	Strict unimodality of the spine weights	11
9	Binary transitions force recurrent large amplitudes	12
10	Proof of the zero-radius theorem	13
11	The inverse-iterate theorem	13
11.1	Interior diffeomorphism structure	13
11.2	Formal-radius invariance under local inversion	14
11.3	Endpoint obstruction and iterated inverse asymptotics	15
11.4	The fixed center: a convergent Taylor series that fails to represent	16
11.5	No local elementary representation	16
11.6	What formal reversion does and does not preserve	16
12	Numerical diagnostics	16
12.1	Sinc-product reconstruction	17
12.2	Forward spines and inverse formal reversion	17
13	Connections to the broader Fabius corpus	20
13.1	Bell polynomials and Bernoulli cumulants	20
13.2	q -binomial and Thue–Morse structure	20
13.3	Infinite sinc products	21
13.4	Lambert- W endpoint asymptotics	21
13.5	Legendre expansions and polynomial representations	21
14	Conjectures and research directions	21
14.1	Promising formalization route	23
14.2	Generalization to other atomic functions	23
15	Summary of the proof mechanism	23
A	Auxiliary details on the partition defect	24

B	Reproducibility guide	24
C	Repository source map	25

1 Statement of the problem and strengthened results

The bounded Fabius function is the unique smooth increasing bijection $F: [0, 1] \rightarrow [0, 1]$ satisfying

$$F(0) = 0, \quad F(1 - x) = 1 - F(x), \quad F'(x) = 2F(2x) \quad (0 \leq x \leq \tfrac{1}{2}), \quad (1)$$

with the usual constant continuation outside $[0, 1]$ when the bounded totalization is used. Put

$$I := F^{-1}, \quad G_n := F^{\circ n}, \quad K_n := I^{\circ n}, \quad G_0 = K_0 = \text{id}. \quad (2)$$

Associativity of composition gives the exact inverse identity

$$K_n = G_n^{-1}, \quad G_n = K_n^{-1}. \quad (3)$$

The inverse is C^∞ on $(0, 1)$, though it has singular endpoint slopes.

Theorem 1.1 (Main theorem: inverse iterates are nowhere analytic). *For every integer $n \geq 1$, the n -fold inverse iterate*

$$K_n = I^{\circ n}$$

is not real analytic at any point of $[0, 1]$.

The proof yields a stronger statement about formal Taylor series. For a smooth function h near x , write

$$R_h(x) := \text{rad} \sum_{m \geq 0} \frac{h^{(m)}(x)}{m!} z^m$$

for the radius of convergence of its formal Taylor series; the series need not represent h even when this radius is positive.

Theorem 1.2 (Co-countable zero-radius set for inverse iterates). *For each fixed $n \geq 1$ there exists a co-countable dense set $Y_n \subset (0, 1)$ such that*

$$R_{K_n}(y) = 0, \quad \limsup_{m \rightarrow \infty} \left(\frac{|K_n^{(m)}(y)|}{m!} \right)^{1/m} = +\infty \quad (y \in Y_n). \quad (4)$$

More precisely, one may take $Y_n = G_n(E_n)$, where the explicit co-countable dense set E_n is defined in (47) from forward orbit arithmetic.

The exceptional points are genuine and instructive.

Theorem 1.3 (Affine but nonrepresenting center jet). *For every $n \geq 1$,*

$$K_n(\tfrac{1}{2}) = \tfrac{1}{2}, \quad K_n'(\tfrac{1}{2}) = 2^{-n}, \quad K_n^{(m)}(\tfrac{1}{2}) = 0 \quad (m \geq 2). \quad (5)$$

Thus the Taylor series of K_n at $1/2$ is an affine polynomial of infinite radius, but it does not agree with K_n on any neighborhood.

The forward theorem needed for the proof is itself stronger than nowhere analyticity.

Theorem 1.4 (Forward engine theorem). *For every integer $n \geq 1$, $G_n = F^{\circ n}$ is not real analytic at any point of $[0, 1]$.*

Theorem 1.5 (Forward co-countable zero-radius set). *Fix $n \geq 1$. There exists a co-countable dense set $E_n \subset (0, 1)$ such that, for every $x \in E_n$,*

$$\limsup_{m \rightarrow \infty} \left(\frac{|G_n^{(m)}(x)|}{m!} \right)^{1/m} = +\infty. \quad (6)$$

Consequently $R_{G_n}(x) = 0$ for every $x \in E_n$.

The logical order of the report is deliberate. We first establish the forward engine by a derivative-spine argument. We then transfer both nonanalyticity and zero Taylor radius through local inversion. The endpoint and center analyses are treated separately because the inverse is not a local diffeomorphism at the endpoints and because the center has a convergent but nonrepresenting Taylor series.

2 Repository audit and novelty boundary

The source audit was organized around the repository’s canonical documents and its own consolidation manifests. This is important because the tree contains live proof-backed expositions, a formal Lean development, canonical frontier syntheses, and many frozen or mechanically merged historical drafts. Treating every duplicate as an independent source would obscure rather than improve provenance.

2.1 Canonical sources inspected

The following live documents form the mathematical backbone of the present report.

Repository document	Role in the present proof
Fabius_Function_and_Rvachev_Up.tex	Primary proof-backed exposition: definitions, symmetry, strict monotonicity, convexity and concavity, the signed global extension, exact all-order derivative formulae, sharp derivative bounds, dyadic jets, inverse-function facts, and the formal base nowhere-analyticity statement.
Fabius_Function_Mathematical_Glossary.tex	Notation and cross-reference audit, including the bounded function, signed global extension, Rvachev bump, dyadic grids, Thue–Morse signs, and analytic-locus language.
Non_Elementarity_of_the_Fabius_Function.tex	Dense analyticity of elementary-function classes, the exact analytic locus of F , and the corresponding obstruction for F^{-1} .
fabius_lean_walkthrough.tex	Module-level proof map for the formal derivative, flatness, inverse, and nowhere-analyticity results.
semi-formalized-research-frontiers.tex	Canonical synthesis of q-binomial, sinc-product, Lambert– W , Bell/Bernoulli, Legendre, inverse, sampling, and asymptotic directions.

The archive index and the semi-formalized-frontier README were used as provenance crosswalks for first versions, absorbed notebooks, mechanically merged draft groups, and editorial syntheses. Thus historical copies were audited through their canonical successors and checksummed consolidation records rather than cited as competing statements. The repository itself explicitly separates proof-backed claims from frontier statements awaiting literal formal counterparts; this report follows the same convention.

2.2 Existing results and the novelty boundary

The repository already proves the exact analytic locus of the bounded function,

$$F \text{ is analytic at } x \iff x \notin [0, 1], \quad (7)$$

and proves the corresponding one-fold statement for $I = F^{-1}$. The inverse proof uses positivity of F' on $(0, 1)$, the real-analytic inverse-function theorem, and a separate endpoint obstruction. These results settle $n = 1$ but do not by themselves settle $n \geq 2$.

There are two nontrivial gaps to bridge. First, nowhere analyticity is not preserved by composition for arbitrary smooth functions: chain-rule cancellation may remove a singularity, and eventual vanishing of one jet does not imply eventual vanishing after composition. Second, even after proving the forward theorem for G_n , one still needs to distinguish three inverse questions: local analytic representability, convergence of the formal inverse Taylor series, and endpoint regularity. The present report addresses all three.

The corrected sibling report `drafts/representations/fabius_iterates_nowhere_analytic/` now proves the all- n forward theorem and already records inverse nowhere analyticity as a corollary. Accordingly, Sections 5–10 are a derived copy of that forward engine, not independent novelty. They are retained here so the inverse argument can be read in one document, with all corrections from the sibling report ported verbatim. The distinct manuscript layer begins in Section 11: formal Taylor-radius transport through smooth local inversion, the affine but nonrepresenting center jet, and the iterated inverse endpoint scale. The corpus formalizes the $n = 1$ inverse analytic locus and one-fold endpoint Hölder obstruction, but it does not yet contain exact Lean counterparts for these $n \geq 2$ iterate statements.

3 Fabius, Rvachev, Thue–Morse, and the derivative atlas

3.1 The Rvachev bump and the signed global extension

Let up denote Rvachev’s compactly supported C^∞ function, normalized by

$$\text{supp } \text{up} = [-1, 1], \quad \text{up}(0) = 1,$$

and related to the bounded Fabius function by

$$\text{up}(t) = F(1 - |t|) \quad (-1 \leq t \leq 1). \quad (8)$$

In particular, up is even, strictly decreasing on $[0, 1]$, and

$$\text{up}(\tfrac{1}{2}) = F(\tfrac{1}{2}) = \tfrac{1}{2}. \quad (9)$$

For $k \geq 0$, put

$$\varepsilon_k = (-1)^{s_2(k)}, \quad (10)$$

where $s_2(k)$ is the sum of the binary digits of k . The signed global extension Θ can be written blockwise as

$$\Theta(t) = \varepsilon_k \text{up}(t - (2k + 1)), \quad 2k \leq t \leq 2k + 2. \quad (11)$$

It agrees with F on $[0, 1]$, satisfies $\Theta'(t) = 2\Theta(2t)$ globally, and obeys

$$|\Theta(t)| \leq 1. \quad (12)$$

The translate signs in (11) are precisely the Thue–Morse signs.

3.2 Exact all-order derivatives

Define the superfactorial dyadic scale

$$Q_m := 2^{\binom{m+1}{2}} = 2^{m(m+1)/2}. \quad (13)$$

Repeated differentiation of the dilation equation gives, for every $m \geq 1$ and $x \in (0, 1)$,

$$F^{(m)}(x) = Q_m \Theta(2^m x). \quad (14)$$

Consequently

$$\left| F^{(m)}(x) \right| \leq Q_m. \quad (15)$$

At the matched dyadic grid this becomes the exact Thue–Morse jet law

$$F^{(m)}\left(\frac{a}{2^m}\right) = \begin{cases} 0, & a \text{ even}, \\ Q_m \varepsilon_{\lfloor a/2 \rfloor}, & a \text{ odd}. \end{cases} \quad (16)$$

The factor Q_m is the source of nonanalytic growth. Since

$$Q_m^{1/m} = 2^{(m+1)/2}, \quad (17)$$

no fixed ordinary exponential can neutralize it after division by $m!$. The only serious issue is cancellation among the many chain-rule terms created by composition.

3.3 Shape and orbit dynamics

The repository proves that F is a strictly increasing bijection of $[0, 1]$, that F' is strictly increasing on $[0, \frac{1}{2}]$ and strictly decreasing on $[\frac{1}{2}, 1]$, and that

$$F'(0) = 0, \quad F'(\tfrac{1}{2}) = 2, \quad F'(1) = 0. \quad (18)$$

Moreover F is strictly convex on $[0, \frac{1}{2}]$ and strictly concave on $[\frac{1}{2}, 1]$. Comparing with the chord joining $(0, 0)$ and $(\frac{1}{2}, \frac{1}{2})$ gives

$$F(x) < x \quad (0 < x < \tfrac{1}{2}), \quad F(x) > x \quad (\tfrac{1}{2} < x < 1). \quad (19)$$

Thus every noncentral interior orbit moves monotonically toward an endpoint:

$$0 < x < \tfrac{1}{2} \implies x > F(x) > F^{\circ 2}(x) > \cdots > 0, \quad (20)$$

$$\tfrac{1}{2} < x < 1 \implies x < F(x) < F^{\circ 2}(x) < \cdots < 1. \quad (21)$$

This elementary dynamical fact will force strict unimodality of the exponential weights in the derivative spine sum.

4 Why a direct dyadic-jet induction fails

It is useful to see the obstruction before introducing the cure. For two smooth functions f and h ,

$$(f \circ h)^{(m)}(x) = \sum_{\pi \in \Pi_m} f^{(|\pi|)}(h(x)) \prod_{B \in \pi} h^{(|B|)}(x), \quad (22)$$

where Π_m is the set of partitions of $\{1, \dots, m\}$. Even if $h^{(r)}(x) = 0$ for all large r , partitions with many small blocks can survive. Similarly, even if the high derivatives of f vanish at $h(x)$, the term $f'(h(x))h^{(m)}(x)$ survives. There is no pointwise “eventually zero jet” induction for $F^{\circ n}$.

At the opposite extreme, a crude absolute-value bound on (22) introduces Bell numbers. Since the number of partitions is superexponential on an ordinary exponential scale, such a bound loses the delicate comparison with Q_m . The correct normalization must exploit the quadratic exponent in Q_m , not merely count partitions. This leads to the defect below.

5 The dyadic defect of a Faà di Bruno partition

Let $\pi \in \Pi_m$ have $k = |\pi|$ blocks of sizes $r_1, \dots, r_k$, so $r_1 + \dots + r_k = m$. Define

$$D(\pi) := \binom{m}{2} - \binom{k}{2} - \sum_{i=1}^k \binom{r_i}{2}. \quad (23)$$

An equivalent and more transparent formula is

$$D(\pi) = \sum_{1 \leq i < j \leq k} (r_i r_j - 1). \quad (24)$$

Every summand is nonnegative. Therefore $D(\pi) = 0$ only in the two extremal cases: there is one block, or every block is a singleton. These are exactly the two unbranched chain-rule spines.

Lean crosswalk: the finite positive-list defect is complete. The module `PartitionDefect.lean` has exactly three public definitions and thirty-three public theorems; the following inventory is exhaustive. Its list $r = [r_1, \dots, r_k]$ records positive block sizes, but it does not construct a set partition of $\{1, \dots, m\}$.

- The three definitions are `pairSum`, `blockPairDefect`, and `partitionDefect`.
- The unordered-pair and triangular-number engine consists of `pairSum_nil`, `pairSum_cons`, `pairSum_one`, `pairSum_add`, `pairSum_congr`, `pairSum_map`, `choose_add_two`, and `choose_list_sum_two`.
- The conventional defect formulas, excess decomposition, and fixed-block lower bound are `blockPairDefect_eq_mul_sub_one`, `partitionDefect_eq_pairSum_mul_sub_one`, `partitionDefect_nonneg`, `choose_sum_two_eq_choose_length_add_sum_add_partitionDefect`, `partitionDefect_eq_choose_sum_sub_choose_length_sub_sum`, `length_le_sum_of_pos`, `sum_map_sub_one`, `pairSum_add_eq`, `pairSum_map_add_eq`, `partitionDefect_eq_linear_add_pairwise_excess`, `partitionDefect_lower_bound`, and `partitionDefect_fixed_block_bound`.
- The zero and sharp-equality classifiers are `pairSum_eq_zero_iff_pairwise`, `blockPairDefect_eq_zero_iff`, `partitionDefect_eq_zero_iff`, `sub_one_mul_sub_one_eq_zero_iff`, `partitionDefect_eq_lower_bound_iff`, and `partitionDefect_fixed_block_eq_iff`.
- The first-positive-shell closure is `add_sub_one_le_mul_of_pos`, `mul_eq_add_sub_one_iff`, `firstShell_le_fixedBlockProduct`, `fixedBlockProduct_eq_firstShell_iff`, `firstShell_le_partitionDefect`, `partitionDefect_eq_firstShell_iff`, and `partitionDefect_twoBlock_firstShell`.

For positive entries the module proves

$$D(r) = (k-1) \sum_i (r_i - 1) + \sum_{i < j} (r_i - 1)(r_j - 1) \geq (k-1)(m-k).$$

It proves $D(r) = 0$ exactly when $k \leq 1$ or every entry is one; equality in the fixed- m , fixed- k bound exactly when every pair contains a singleton (equivalently at most one entry exceeds one); and, under precisely $2 \leq k < m$, the first-shell bound $D(r) \geq m - 2$, with equality exactly at $k = 2$ or $k = m - 1$ together with that same one-nonsingleton condition. The explicit profile $[m - 1, 1]$ attains the shell for $m \geq 3$.

This closure does *not* formalize the set-partition family Π_m , the quadratic-scale factorization (26), the weighted Bell sum (27), its asymptotics, the two-spine reduction, the finite spine expansion, or either the forward- or inverse-iterate theorem. Those steps remain paper-only.

Let $u(\pi)$ be the number of singleton blocks and let

$$s(\pi) := k - u(\pi) \quad (25)$$

be the number of non-singleton blocks.

Lemma 5.1 (Quadratic-scale factorization). *For every partition $\pi \in \Pi_m$,*

$$\frac{Q_k \prod_{r_i \geq 2} Q_{r_i}}{Q_m} = 2^{s(\pi) - D(\pi)}. \quad (26)$$

Proof. Using $\log_2 Q_r = r(r+1)/2$ and omitting singleton blocks from the product,

$$\begin{aligned} 2 \log_2 \frac{Q_k \prod_{r_i \geq 2} Q_{r_i}}{Q_m} &= k(k+1) + \sum_{r_i \geq 2} r_i(r_i+1) - m(m+1) \\ &= k^2 + s(\pi) + \sum_{r_i \geq 2} r_i^2 - m^2. \end{aligned}$$

On the other hand, expanding (23) gives

$$2D(\pi) = m^2 - k^2 + s(\pi) - \sum_{r_i \geq 2} r_i^2.$$

Subtracting proves (26). $\square$

The next proposition is the combinatorial engine. Its role is stronger than saying that each nonextremal partition has positive defect: it shows that the sum over all of them is still exponentially small.

Proposition 5.2 (Weighted defect decay). *For each fixed $K > 0$ there exist constants $C_K > 0$ and $\rho_K \in (0, 1)$ such that*

$$\sum_{\substack{\pi \in \Pi_m \\ 1 < |\pi| < m}} (2K)^{s(\pi)} 2^{-D(\pi)} \leq C_K \rho_K^m \quad (m \geq 2). \quad (27)$$

Proof. We give the full counting argument because this is the point at which a naive Bell number estimate fails.

Write $k = |\pi|$, let

$$e := m - k \quad (28)$$

be the total block excess over singletons, and let $r_{\max} = m - t$ be the largest block size. Convexity of $r \mapsto \binom{r}{2}$ shows that, for fixed m and k , the defect is minimized by one block of size $m - k + 1$ and $k - 1$ singleton blocks. Hence

$$D(\pi) \geq e(k-1) = e(m-e-1). \quad (29)$$

The contribution between a largest block and all other blocks gives the second bound

$$D(\pi) \geq (m-t)t - (k-1) \geq t(m-t-1). \quad (30)$$

First consider partitions with $1 \leq e \leq m/4$. Their non-singleton blocks contain at most $2e$ elements, because the number of such blocks is at most e . Choosing those elements and partitioning them gives at most

$$(2e+1)m^{2e}(2e)^{2e} \quad (31)$$

possibilities. Also $s(\pi) \leq e$, while (29) gives $D(\pi) \geq e(3m/4 - 1)$. Uniformly for $1 \leq e \leq m/4$, the base-two logarithm of the counting and weight factor is at most

$$\log_2(2e+1) + 4e \log_2 m + e \log_2 \max(1, 2K).$$

For fixed K this is at most $em/4$ once m is large enough, whereas the negative defect exponent is at most $-e(3m/4 - 1)$. Thus the e -th contribution is at most $2^{-em/3}$ after enlarging the threshold once more, uniformly in e . The resulting geometric sum is $\mathcal{O}(2^{-m/3})$.

Next consider partitions with $1 \leq t \leq m/4$. Choose the t elements outside a largest block and partition them. Up to harmless overcounting there are at most m^{tt} choices, and $s(\pi) \leq t + 1$. By (30), $D(\pi) \geq t(3m/4 - 1)$. The same explicit estimate as above, now with t in place of e (and one harmless extra factor $\max(1, 2K)$), bounds the t -th contribution by $2^{-tm/3}$ for all sufficiently large m , uniformly for $1 \leq t \leq m/4$. Its sum is again exponentially small.

It remains to treat the middle region $e > m/4$ and $t > m/4$. If $t \leq 3m/4$, then (30) gives $D(\pi) \geq c_1 m^2$ for all sufficiently large m and an absolute $c_1 > 0$. Suppose instead that $t > 3m/4$, so every block has size less than $m/4$. If $k > m/4$, then (29) gives $D(\pi) \geq c_2 m^2$. If $k \leq m/4$, then

$$\begin{aligned} D(\pi) &= \sum_{i < j} r_i r_j - \binom{k}{2} \\ &= \frac{m^2 - \sum_i r_i^2}{2} - \binom{k}{2} > \frac{3m^2}{8} - \frac{m^2}{32} = \frac{11m^2}{32}. \end{aligned}$$

In the middle region, the total number of partitions is at most m^m and the weight $(2K)^{s(\pi)}$ is at most $\max(1, 2K)^m$. Both are overwhelmed by 2^{-cm^2} . Combining the three regions proves (27), after increasing C_K to cover the finitely many small values of m . $\square$

6 A two-spine composition lemma

The weighted defect estimate converts Faà di Bruno's formula into an asymptotically binary chain rule.

Lemma 6.1 (Two-spine reduction). *Let h be smooth near x , let f be smooth near $y = h(x)$, and put $B = |h'(x)|$. Suppose there are constants $H > 0$ and $K \geq 1$ such that*

$$\left| f^{(r)}(y) \right| \leq Q_r \quad (r \geq 1), \quad (32)$$

$$\left| h^{(r)}(x) \right| \leq K Q_r H^r \quad (r \geq 2). \quad (33)$$

Then, with $L = \max(B, H)$,

$$(f \circ h)^{(m)}(x) = f'(y)h^{(m)}(x) + f^{(m)}(y)h'(x)^m + o(Q_m L^m) \quad (m \rightarrow \infty). \quad (34)$$

Proof. In (22), remove the one-block partition and the all-singleton partition. They give exactly the two displayed terms. For any remaining partition π with k blocks, u singleton blocks, and $s = k - u$ non-singleton blocks, the corresponding term has absolute value at most

$$Q_k B^u \prod_{r_i \geq 2} (K Q_{r_i} H^{r_i}) \leq Q_m L^m (2K)^s 2^{-D(\pi)},$$

where lemma 5.1 was used in the last step. Summing and applying proposition 5.2 bounds the entire remainder by $C_K Q_m L^m \rho_K^m = o(Q_m L^m)$. $\square$

Remark 6.2 (Why this lemma is composition-stable). The estimate does not require analyticity, Gevrey bounds of factorial type, or a closed formula for every Bell polynomial. It uses only the exact quadratic dyadic scale Q_r and an ordinary exponential envelope H^r for the inner derivatives. The conclusion has the same form, so it can be iterated a fixed number of times.

7 The orbit-spine asymptotic for $F^{\circ n}$

Fix $x \in (0, 1)$ and $n \geq 1$. Define the finite orbit

$$x_j := G_j(x) = F^{\circ j}(x), \quad 0 \leq j \leq n-1, \quad (35)$$

and the local slopes

$$a_j := F'(x_j) > 0. \quad (36)$$

The prefix and suffix weights are

$$B_0 := 1, \quad B_j := \prod_{r=0}^{j-1} a_r = G'_j(x) \quad (j \geq 1), \quad (37)$$

$$A_{j,n} := \prod_{r=j+1}^{n-1} a_r, \quad 0 \leq j < n, \quad (38)$$

where empty products equal 1. Finally set

$$H_n(x) := \max_{0 \leq j < n} B_j. \quad (39)$$

Theorem 7.1 (Finite spine expansion). *For each fixed $n \geq 1$ and $x \in (0, 1)$,*

$$G_n^{(m)}(x) = Q_m \left[\sum_{j=0}^{n-1} A_{j,n} B_j^m \Theta(2^m x_j) + o(H_n(x)^m) \right] \quad (m \rightarrow \infty). \quad (40)$$

Proof. We induct on n . For $n = 1$, $x_0 = x$, $A_{0,1} = B_0 = H_1 = 1$, and (40) is exactly (14).

Assume the formula for G_j . It immediately supplies a global-in-order envelope at the fixed point x : there is $K \geq 1$ such that

$$\left| G_j^{(r)}(x) \right| \leq K Q_r H_j(x)^r \quad (r \geq 2). \quad (41)$$

Indeed the asymptotic gives the estimate for all sufficiently large r , and K can be enlarged to absorb the finitely many remaining orders.

Apply lemma 6.1 with $f = F$ and $h = G_j$. Since $h(x) = x_j$, $h'(x) = B_j$, $f'(x_j) = a_j$, and $f^{(m)}(x_j) = Q_m \Theta(2^m x_j)$, we obtain

$$G_{j+1}^{(m)}(x) = a_j G_j^{(m)}(x) + Q_m B_j^m \Theta(2^m x_j) + o(Q_m H_{j+1}(x)^m), \quad (42)$$

where $H_{j+1} = \max(H_j, B_j)$. Multiplication by a_j changes every old suffix factor $A_{i,j}$ into $A_{i,j+1}$, while the new term has suffix factor $A_{j,j+1} = 1$. This is precisely (40) for $j+1$. $\square$

Remark 7.2 (Combinatorial interpretation). Each summand in (40) corresponds to a chain-rule tree with one high-valence vertex, located at orbit level j , and unary vertices everywhere else. All trees with two or more genuinely branching vertices are swallowed by the $o(H_n^m)$ remainder. This is why “spine” is an apt name.

8 Strict unimodality of the spine weights

The finite sum in (40) could still cancel if several B_j have the same maximal exponential size. The orbit dynamics show that this is exceptional.

Lemma 8.1 (Decreasing slope sequence). *Let $x \in (0, 1) \setminus \{\frac{1}{2}\}$. Then the sequence*

$$a_0, a_1, a_2, \dots \quad (43)$$

is strictly decreasing.

Proof. If $x < \frac{1}{2}$, then (20) says $x_{j+1} < x_j < \frac{1}{2}$. Since F' is strictly increasing on the left half, $a_{j+1} = F'(x_{j+1}) < F'(x_j) = a_j$. If $x > \frac{1}{2}$, then $x_{j+1} > x_j > \frac{1}{2}$ and F' is strictly decreasing on the right half, giving the same conclusion. $\square$

Lemma 8.2 (Unique or adjacent double maximum). *For $x \in (0, 1) \setminus \{\frac{1}{2}\}$, the finite positive sequence $B_0, \dots, B_{n-1}$ is strictly log-concave in the ratio sense:*

$$\frac{B_{j+1}}{B_j} = a_j \quad \text{is strictly decreasing in } j. \quad (44)$$

Hence its maximum is unique unless $a_k = 1$ for some $0 \leq k < n - 1$; in that case the only maximizers are the adjacent pair $B_k = B_{k+1}$.

Proof. Equation (44) follows from (37) and lemma 8.1. A positive finite sequence whose successive ratios strictly decrease first rises while the ratio exceeds 1, possibly has one flat step when the ratio equals 1, and then falls. The conclusion follows. $\square$

Because F' rises strictly from 0 to 2 on $[0, \frac{1}{2}]$, there is a unique $u \in (0, \frac{1}{2})$ with $F'(u) = 1$. Symmetry gives the only other solution $1 - u$. For a fixed n , the tie set is therefore

$$T_n := \bigcup_{\substack{k \geq 0 \\ k+1 < n}} G_k^{-1} \{u, 1 - u\}. \quad (45)$$

Thus $T_1 = \emptyset$. Each G_k is a homeomorphism of $[0, 1]$, so in general T_n is finite, with at most $2(n - 1)$ points.

9 Binary transitions force recurrent large amplitudes

Let $\mathcal{D}_2$ denote the dyadic rationals in $(0, 1)$.

Lemma 9.1 (Binary-transition lemma). *If $y \in (0, 1) \setminus \mathcal{D}_2$, then there are infinitely many $m \geq 1$ such that*

$$|\Theta(2^m y)| \geq \frac{1}{2}. \quad (46)$$

Proof. Choose the nonterminating binary expansion

$$y = 0.b_1 b_2 b_3 \dots \quad (b_j \in \{0, 1\}).$$

Since y is not dyadic, its digits are not eventually constant: eventual zeros give a terminating expansion, while eventual ones give the alternative expansion of a dyadic rational. Hence there are infinitely many transitions $b_m \neq b_{m+1}$.

Modulo 2, the number $2^m y$ is

$$b_m + 0.b_{m+1} b_{m+2} \dots$$

At a transition this lies in $[\frac{1}{2}, \frac{3}{2}]$. By the block formula (11), the centered argument of up then lies in $[-\frac{1}{2}, \frac{1}{2}]$. Since up is even and decreasing on $[0, 1]$, (9) gives

$$|\Theta(2^m y)| = \text{up}(\text{centered argument}) \geq \text{up}(\frac{1}{2}) = \frac{1}{2}.$$

There are infinitely many such m . $\square$

Remark 9.2. The lemma uses only adjacent binary digit changes, not equidistribution or normality. It therefore applies to every non-dyadic point, including highly non-generic automatic, Sturmian, or sparse-digit numbers.

10 Proof of the zero-radius theorem

For fixed n , define

$$E_n := (0, 1) \setminus \left(\left\{ \frac{1}{2} \right\} \cup T_n \cup \bigcup_{j=0}^{n-1} G_j^{-1}(\mathcal{D}_2) \right). \quad (47)$$

Every G_j is a homeomorphism, so each preimage $G_j^{-1}(\mathcal{D}_2)$ is countable. The tie set T_n is finite. Thus E_n is co-countable and dense in $(0, 1)$.

Proposition 10.1 (Dominant-spine lower bound). *Let $x \in E_n$, and let j_* be the unique index for which $B_{j_*} = H_n(x)$. There are infinitely many m such that*

$$\left| G_n^{(m)}(x) \right| \geq c(x, n) Q_m H_n(x)^m \quad (48)$$

with a constant $c(x, n) > 0$ independent of m .

Proof. Because $x \in E_n$, the orbit point x_{j_*} is not dyadic. By lemma 9.1, there is an infinite set $M \subset \mathbb{N}$ such that

$$|\Theta(2^m x_{j_*})| \geq \frac{1}{2} \quad (m \in M). \quad (49)$$

For every $j \neq j_*$, put $r_j = B_j/H_n(x) < 1$. Divide (40) by $Q_m H_n(x)^m$. Along M , the dominant summand has absolute value at least $A_{j_*,n}/2$, while the sum of all other spines is bounded by

$$\sum_{j \neq j_*} A_{j,n} r_j^m \rightarrow 0.$$

The normalized remainder also tends to zero. Therefore, for all sufficiently large $m \in M$, the normalized absolute value is at least $A_{j_*,n}/4$. We may take $c(x, n) = A_{j_*,n}/4 > 0$. $\square$

Proof of theorem 1.5. At every $x \in E_n$, proposition 10.1 and $m! \leq m^m$ give along an infinite subsequence

$$\left(\frac{|G_n^{(m)}(x)|}{m!} \right)^{1/m} \geq c(x, n)^{1/m} H_n(x) \frac{Q_m^{1/m}}{(m!)^{1/m}} \quad (50)$$

$$\geq c(x, n)^{1/m} H_n(x) \frac{2^{(m+1)/2}}{m} \rightarrow +\infty. \quad (51)$$

This proves (6). The Cauchy–Hadamard formula then gives zero radius of convergence for the Taylor series at x . $\square$

Proof of theorem 1.4. Suppose G_n were real analytic at some $p \in [0, 1]$. Real analyticity is a local open property: there would be a radius $r > 0$ such that G_n is represented by a convergent power series throughout $(p - r, p + r)$, and hence is analytic at every point of that interval. The interval intersects $(0, 1)$ in a nonempty open set, even when p is an endpoint. Since E_n is dense, choose $x \in E_n \cap (p - r, p + r)$. This contradicts the zero Taylor radius at x established in theorem 1.5. $\square$

11 The inverse-iterate theorem

11.1 Interior diffeomorphism structure

Let $I = F^{-1}$. Since $F'(x) > 0$ for $0 < x < 1$, every G_n is a strictly increasing C^∞ diffeomorphism of $(0, 1)$ onto itself, and its inverse is K_n . For $y \in (0, 1)$ and $x = K_n(y)$,

$$K'_n(y) = \frac{1}{G'_n(x)} = \prod_{j=0}^{n-1} \frac{1}{F'(F^{\circ j}(x))} > 0. \quad (52)$$

Equivalently, writing $y_j = I^{\circ j}(y)$,

$$K'_n(y) = \prod_{j=0}^{n-1} I'(y_j). \quad (53)$$

First proof of [theorem 1.1](#) on $(0, 1)$. Suppose K_n were real analytic at some $y_0 \in (0, 1)$. Its derivative is nonzero by [\(52\)](#). The real-analytic inverse-function theorem then makes its local inverse G_n analytic at $x_0 = K_n(y_0)$, contradicting [theorem 1.4](#). $\square$

This short argument proves local nonrepresentability, but it does not yet say whether the formal Taylor series of K_n converges to some unrelated analytic germ. Formal reversion settles that stronger question.

11.2 Formal-radius invariance under local inversion

Theorem 11.1 (Convergence is invariant under formal reversion). *Let g and k be C^∞ local inverses near x_0 and $y_0 = g(x_0)$, with $g'(x_0) \neq 0$. Then*

$$R_g(x_0) > 0 \iff R_k(y_0) > 0. \quad (54)$$

Consequently $R_g(x_0) = 0$ if and only if $R_k(y_0) = 0$.

Proof. Translate the centers and write the formal Taylor series as

$$\widehat{g}(z) = a_1 z + a_2 z^2 + \cdots, \quad \widehat{k}(w) = b_1 w + b_2 w^2 + \cdots, \quad a_1 b_1 = 1.$$

Differentiating the smooth identities $g \circ k = \text{id}$ and $k \circ g = \text{id}$ to every finite order gives the formal composition identities

$$\widehat{g} \circ \widehat{k} = w, \quad \widehat{k} \circ \widehat{g} = z. \quad (55)$$

Thus each is the unique formal compositional inverse of the other.

Assume $R_k(y_0) > 0$. Then $\widehat{k}$ defines an analytic germ with nonzero linear coefficient. By the analytic inverse-function theorem it has a convergent analytic inverse germ. The Taylor series of that inverse is the unique formal inverse of $\widehat{k}$, hence equals $\widehat{g}$. Therefore $R_g(x_0) > 0$. Interchanging g and k proves the converse. $\square$

Corollary 11.2 (Exact transport of the Taylor-radius locus). *For every $n \geq 1$ and $x \in (0, 1)$,*

$$R_{K_n}(G_n(x)) = 0 \iff R_{G_n}(x) = 0. \quad (56)$$

The same equivalence holds with “ $= 0$ ” replaced by “ > 0 ”.

Proof of [theorem 1.2](#). Let E_n be the co-countable dense set in [\(47\)](#) and define

$$Y_n := G_n(E_n). \quad (57)$$

Because G_n is a homeomorphism, Y_n is dense. Its complement is the image under G_n of the countable set $(0, 1) \setminus E_n$, so Y_n is co-countable. If $y = G_n(x)$ with $x \in E_n$, then $R_{G_n}(x) = 0$ by [theorem 1.5](#); [theorem 11.1](#) gives $R_{K_n}(y) = 0$. The Cauchy–Hadamard formula is exactly the limsup assertion in [\(4\)](#). $\square$

The theorem gives a second proof of interior nowhere analyticity: an analytic point has an open analytic neighborhood, but every nonempty open interval meets the dense set Y_n on which the Taylor radius is zero.

11.3 Endpoint obstruction and iterated inverse asymptotics

There is an elementary endpoint obstruction even before using the sharp asymptotic. Since $F'(0) = 0$,

$$\frac{F(x)}{x} \longrightarrow 0 \quad (x \downarrow 0). \quad (58)$$

Writing $y = F(x)$ gives

$$\frac{I(y)}{y} = \frac{x}{F(x)} \longrightarrow +\infty. \quad (59)$$

Moreover $F(x) < x$ on $(0, 1/2)$, so $I(y) > y$ there and therefore $K_n(y) \geq I(y)$. Hence

$$\frac{K_n(y)}{y} \longrightarrow +\infty. \quad (60)$$

Thus K_n has no finite right derivative at 0 and cannot be analytic there. The symmetry $I(1 - y) = 1 - I(y)$, inherited by every iterate, gives the same conclusion at 1. Together with the interior proof, this completes [theorem 1.1](#) under either the one-sided or totalized endpoint convention.

The repository's formally backed endpoint theorem yields substantially more. Put $L = \log 2$. Its quadratic logarithmic law is

$$-\log F(x) \sim \frac{(\log(1/x))^2}{2L} \quad (x \downarrow 0). \quad (61)$$

Theorem 11.3 (Iterated inverse endpoint scale). *For each fixed $n \geq 1$, as $y \downarrow 0$,*

$$-\log K_n(y) \sim c_n (\log(1/y))^{2^{-n}}, \quad c_n = (2 \log 2)^{1-2^{-n}}. \quad (62)$$

Consequently, for every $\alpha > 0$,

$$\frac{K_n(y)}{y^\alpha} \longrightarrow +\infty. \quad (63)$$

Thus K_n is not Hölder continuous of any positive order at either endpoint.

Proof. For $y \downarrow 0$, let $T_0 = \log(1/y)$ and recursively put $T_j = \log(1/K_j(y))$. If $u = I(v)$, then $v = F(u)$, so (61) gives

$$\log(1/u) \sim \sqrt{2L \log(1/v)}. \quad (64)$$

Hence $T_{j+1} \sim \sqrt{2LT_j}$. Induction gives

$$T_n \sim c_n T_0^{2^{-n}}, \quad c_{n+1} = \sqrt{2Lc_n}, \quad c_0 = 1.$$

Solving the constant recursion yields $c_n = (2L)^{1/2+1/4+\dots+1/2^n} = (2L)^{1-2^{-n}}$, proving (62). Since $T_n = o(T_0)$,

$$\log \frac{K_n(y)}{y^\alpha} = \alpha T_0 - T_n \longrightarrow +\infty,$$

which proves (63). Reflection gives the endpoint 1. $\square$

The sharper repository expansion inserts a lower-branch Lambert- W phase and a tiny period-one ripple. Iterating that full expansion should produce a hierarchy of nested Lambert phases; the coarse law above is the exact leading scale and is all that is needed for the regularity theorem.

11.4 The fixed center: a convergent Taylor series that fails to represent

Proof of theorem 1.3. The exact derivative atlas gives

$$F(\tfrac{1}{2}) = \tfrac{1}{2}, \quad F'(\tfrac{1}{2}) = 2, \quad F^{(m)}(\tfrac{1}{2}) = 0 \quad (m \geq 2).$$

Formal composition therefore gives

$$\widehat{G}_n(\tfrac{1}{2} + h) = \tfrac{1}{2} + 2^n h.$$

Its unique formal inverse is $\tfrac{1}{2} + 2^{-n}h$, proving the derivative identities for K_n . If this Taylor polynomial represented K_n on a neighborhood, then K_n would be analytic there, contradicting theorem 1.1. Thus the entire nonlinear remainder is flat at the center but nonzero on every punctured neighborhood. $\square$

11.5 No local elementary representation

The repository's non-elementarity development proves that every function in its formal elementary real-variable class has a dense open analytic locus.

Corollary 11.4 (Local non-elementarity). *Fix $n \geq 1$. No elementary function in the repository's class can agree with $K_n = I^{\circ n}$ on a subset of $(0, 1)$ having nonempty interior. The same statement holds for $G_n = F^{\circ n}$.*

Proof. Agreement on an open interval with a function having a dense analytic locus would make the corresponding iterate analytic at some point of that interval, contrary to theorems 1.1 and 1.4. $\square$

11.6 What formal reversion does and does not preserve

The forward lower bound on E_n has the explicit superfactorial scale

$$|G_n^{(m)}(x)| \geq cQ_m H_n(x)^m \quad \text{for infinitely many } m, \quad Q_m = 2^{m(m+1)/2}. \quad (65)$$

Formal reversion proves zero radius for the inverse Taylor series, but by itself it does not identify a comparable pointwise leading term for $K_n^{(m)}$. The inverse Bell polynomials contain additional cancellation. A direct inverse-spine asymptotic is therefore a genuine next problem, not an automatic corollary. The numerical section suggests that the inverse Taylor roots grow even faster at the chosen diagnostic point, but no asymptotic claim is based on those finite-order data.

12 Numerical diagnostics

No numerical experiment is used in any proof. The accompanying commented script `numerical_experiments.py` performs four reproducible checks:

- (i) it reconstructs up by an inverse FFT of the truncated sinc product;
- (ii) it composes centered Taylor series to evaluate the full derivatives of G_n ;
- (iii) it compares those derivatives with the finite spine sum in (40);
- (iv) it reverts the forward Taylor series coefficient by coefficient, using 160-digit arithmetic, to obtain the formal Taylor series of K_n and verify the composition residual.

12.1 Sinc-product reconstruction

With the Fourier convention $\hat{f}(\xi) = \int_{\mathbb{R}} f(t) e^{-2\pi i \xi t} dt$, the Rvachev bump has product

$$\widehat{\text{up}}(\xi) = \prod_{j=0}^{\infty} \text{sinc}\left(\frac{\pi \xi}{2^j}\right), \quad \text{sinc}(z) = \frac{\sin z}{z}. \quad (66)$$

The script truncates the product, applies an inverse FFT on a period larger than the support, normalizes at $\text{up}(0) = 1$, and uses shape-preserving interpolation. It then recovers $F(x) = \text{up}(x - 1)$ on $(0, 1)$ and the signed blocks in (11).

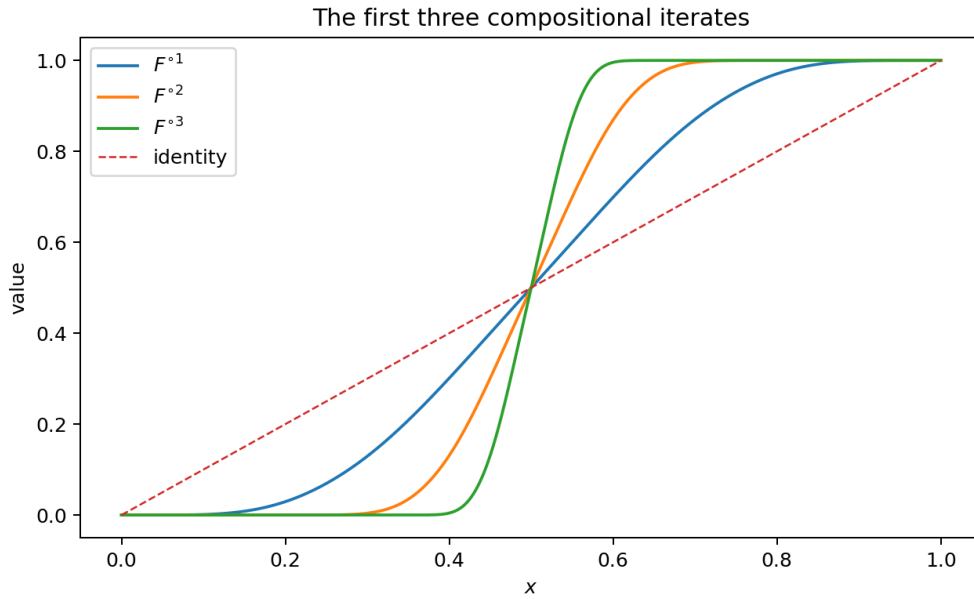


Figure 1: Numerical reconstructions of the first three forward compositional iterates. The strict drift away from the repelling fixed point $1/2$ and toward the endpoints is visible. The inverse iterates reverse these curves across the diagonal.

12.2 Forward spines and inverse formal reversion

The default diagnostic uses

$$x_0 = 0.437123456789, \quad n = 4, \quad y_0 = G_4(x_0) \approx 1.6039811710 \times 10^{-4}.$$

The computed orbit and prefix weights are

$$\begin{aligned} (x_0, x_1, x_2, x_3) &\approx (0.4371234568, 0.3743185303, 0.2522048636, 0.07166875375), \\ (B_0, B_1, B_2, B_3) &\approx (1, 1.992844087, 3.703442407, 3.768767090). \end{aligned}$$

Thus $j = 3$ is the unique dominant forward spine. Let D_m be the fully composed forward derivative and S_m the spine sum. The last two columns below are the m -th roots of the absolute forward and inverse Taylor coefficients.

The script writes all raw values to `numerical_output/spine_diagnostic.csv`, records orbit metadata and the maximum formal reversion residual, and regenerates every figure. At degree 22, using 160 decimal digits, the maximum coefficient residual in $\hat{G}_4 \circ \hat{K}_4 = \text{id}$ is below 2×10^{-65} .

m	$D_m/(Q_m H_4^m)$	$S_m/(Q_m H_4^m)$	$ [z^m]G_4 ^{1/m}$	$ [w^m]K_4 ^{1/m}$
10	1.049415	0.721343	37.84	2.706×10^3
12	-0.503489	-0.378637	60.91	3.059×10^3
13	0.961480	0.998158	84.86	3.209×10^3
15	-0.384836	-0.384202	140.94	3.572×10^3
16	-1.008935	-1.002630	200.75	4.727×10^3
18	0.432079	0.432434	344.83	7.989×10^3
19	-0.998626	-0.999888	486.69	1.118×10^4
21	0.091358	0.091354	793.50	1.795×10^4
22	0.581063	0.581018	1176.18	2.644×10^4

Table 2: Finite-order diagnostics. Oscillation is expected. The forward theorem concerns the normalized asymptotic remainder, while the inverse column only illustrates formal reversion at finite order.

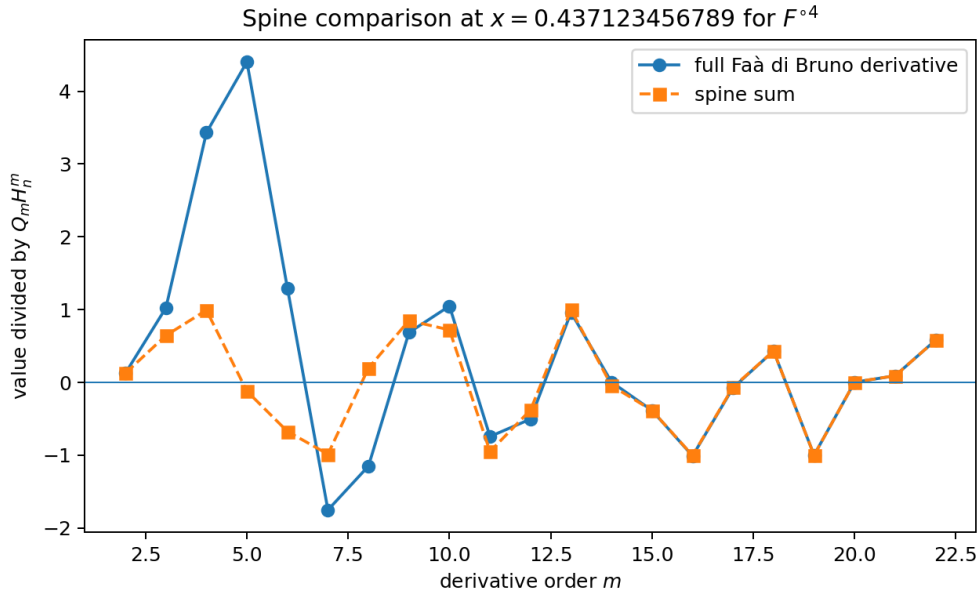


Figure 2: The full forward derivative and the finite spine sum after normalization by $Q_m H_n^m$.

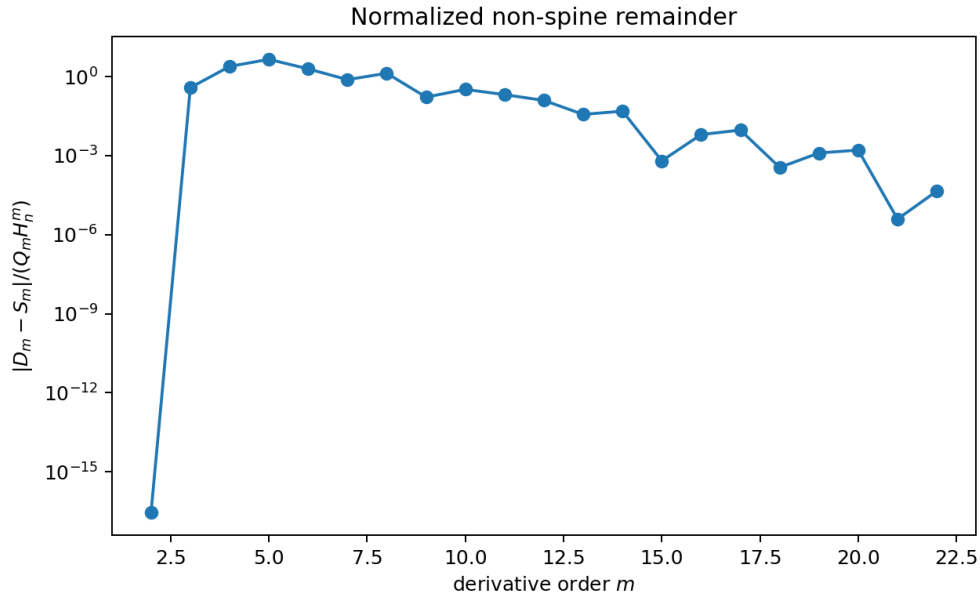


Figure 3: The normalized non-spine remainder. The high-order values illustrate the defect-driven suppression proved in [proposition 5.2](#).

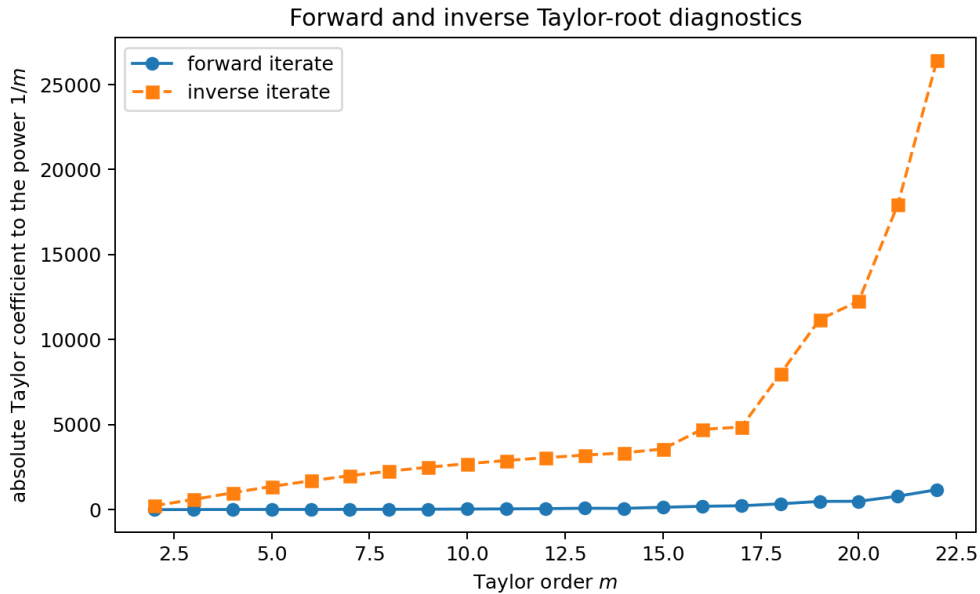


Figure 4: Forward and inverse Taylor-coefficient roots at the paired points x_0 and $y_0 = G_4(x_0)$. The inverse coefficients are obtained by arbitrary-precision formal reversion.

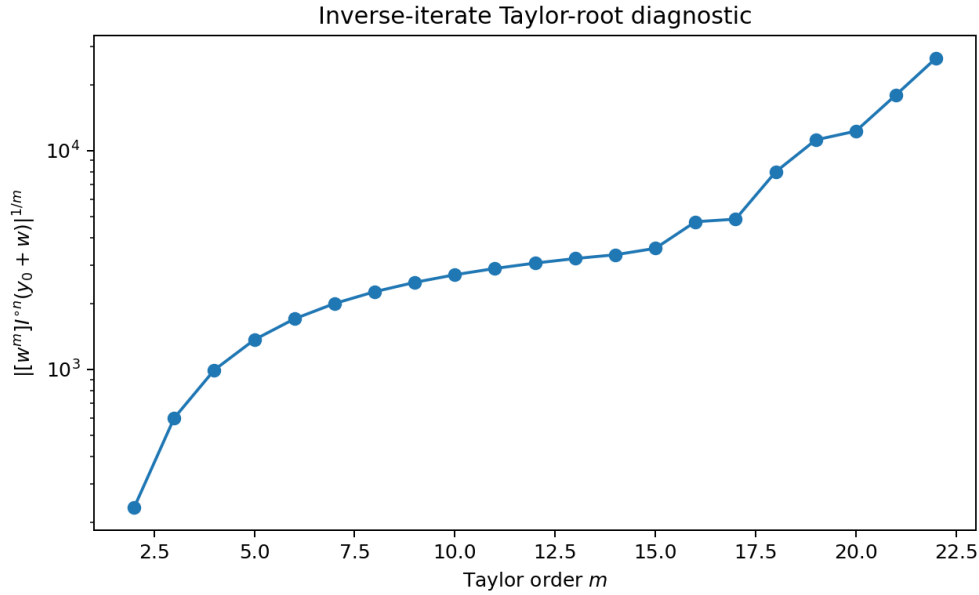


Figure 5: The inverse-iterate Taylor-root diagnostic alone. The values are not used to infer an asymptotic; they audit the coefficient-reversion implementation and illustrate the zero-radius phenomenon proved abstractly.

13 Connections to the broader Fabius corpus

13.1 Bell polynomials and Bernoulli cumulants

Bell polynomials already occur in the repository through moment and cumulant formulae for dyadic values. Here the same combinatorial species—set partitions—appears in a different role: it organizes the derivatives of a composition. The defect $D(\pi)$ adds a quadratic dyadic energy to a Bell partition. This suggests studying the weighted partition polynomial

$$\mathcal{P}_m(z, w) := \sum_{\pi \in \Pi_m} z^{s(\pi)} w^{D(\pi)}. \quad (67)$$

At $w = 1$ this collapses toward familiar Bell refinements; at $w = 1/2$ it is the exact suppression factor governing the present proof. The Bernoulli-cumulant formulas for up arise from logarithms of dyadic products, while (67) measures the combinatorics of composing the resulting quadratic-scale jets. A q -Bell theory with $q = 1/2$ may unify these two appearances.

13.2 q -binomial and Thue–Morse structure

The exact dyadic values of F are governed by Gaussian-binomial and q -Pochhammer identities, whereas the derivative signs in (16) are Thue–Morse. The present proof shows that under finite self-composition these structures do not spread across all Faà di Bruno trees at leading order. They remain concentrated on a finite set of orbit spines. The leading sign-amplitude process is

$$m \mapsto \Theta(2^m x_{j_*}), \quad (68)$$

which is a direct symbolic sampling of the dyadic shift. Thus a high-dimensional Bell expansion asymptotically collapses to a one-dimensional Thue–Morse-modulated orbit observable.

13.3 Infinite sinc products

The sinc product (66) supplies the numerical representation used here and explains the exceptional smoothness of up . The nowhere-analyticity mechanism, however, is most transparent in physical space: dilation creates Q_m , block translation creates Thue–Morse signs, and binary transitions prevent the amplitude from becoming too small forever. The Fourier and derivative pictures are therefore complementary. A possible future goal is to derive the spine expansion directly from a multidimensional Fourier integral for $F^{\circ n}$, but nonlinear composition makes such a representation substantially less direct than the set-partition proof.

13.4 Lambert– W endpoint asymptotics

The repository develops the small-argument asymptotic of F in a phase variable encoded by the lower real branch W_{-1} . Inverting once turns the quadratic logarithmic scale into a square-root logarithmic scale; iterating n times repeatedly halves the logarithmic exponent, as proved in [theorem 11.3](#). The full expansion is subtler: its bounded period-one phase survives inversion and should be evaluated at a nested sequence of Lambert phases. This produces a natural hierarchy

$$T \mapsto \sqrt{2(\log 2)T} \mapsto \sqrt{2(\log 2)\sqrt{2(\log 2)T}} \mapsto \cdots,$$

with lower-order logarithms and periodic corrections at every level. A uniform all-orders theorem for this nested hierarchy would refine the endpoint part of the present report and could quantify the blow-up of K'_n and its higher derivatives.

13.5 Legendre expansions and polynomial representations

The Rvachev function admits rapidly convergent orthogonal-polynomial expansions, and the repository develops finite representations of polynomials by shifted copies of up . Forward iterates become increasingly step-like about $1/2$ as n grows, while inverse iterates become increasingly flat in the center and increasingly steep near the endpoints. Their Legendre coefficients should therefore have complementary two-scale regimes. Nowhere analyticity rules out geometric coefficient decay strong enough to produce a Bernstein-ellipse extension across any real subinterval. The endpoint law (62) suggests that the inverse coefficients may be controlled by stretched-logarithmic saddle points rather than an ordinary algebraic or exponential scale. Determining that scale would connect local inverse singularity to global polynomial approximation.

14 Conjectures and research directions

The main theorem is complete, but it leaves a structured exceptional countable set on which the formal Taylor series may converge without representing the function. The formal-radius transport theorem shows that the inverse classification is exactly the homeomorphic image of the forward positive-radius locus. Of the five submitted conjecture environments, only two remain live manuscript conjectures. Three are retained as explicit warning records: one has a false inverse clause, one is already proved from exact quarter-point facts, and one duplicates the corrected forward report’s sole live conjecture. Neither live conjecture has an exact proved Lean counterpart.

Warning 14.1 (Former Conjecture 14.1 has a false inverse clause). The submitted statement first proposed that the positive-radius locus of G_n is precisely its eventually-zero-jet locus, then claimed that [theorem 11.1](#) gives the same eventual-zero classification for K_n . The second clause is false: formal reversion preserves convergence, not polynomiality. At $x = 1/4$ the exact forward Taylor germ is

$$\widehat{F}_{1/4}(z) = \frac{5}{72} + z + 4z^2,$$

because $F'(1/4) = 1$, $F''(1/4) = 8$, and all higher derivatives vanish. At $y = F(1/4) = 5/72$, the inverse Taylor germ therefore solves $w = z + 4z^2$ and is

$$z = \frac{\sqrt{1 + 16w} - 1}{8},$$

an infinite convergent series. Thus $K_1 = I$ has positive Taylor radius at $5/72$ without an eventually-zero jet. The forward-only assertion remains a plausible open classification problem, especially for $n \geq 2$, but no inverse eventual-zero equivalence is asserted here.

The center is exceptional in both directions because its forward and inverse formal series are affine. At noncentral dyadic points for $n = 1$, however, finite nonlinear forward Taylor polynomials revert to infinite convergent inverse series, exactly as the quarter-point example shows.

Warning 14.2 (Former Conjecture 14.2 is already discharged). The submitted tie-cancellation statement is not a live conjecture. If $x \in T_n$, the tied adjacent weights satisfy $a_k = 1$. Strict monotonicity of F' , together with $F'(1/4) = F'(3/4) = 1$, forces $x_k \in \{1/4, 3/4\}$. Hence $x_{k+1} = F(x_k) \in \{5/72, 67/72\}$. The first orbit point is dyadic, so $\Theta(2^m x_k) = 0$ for every sufficiently large m by (16); the second is non-dyadic, so [lemma 9.1](#) supplies infinitely many m with $|\Theta(2^m x_{k+1})| \geq 1/2$. Since the fixed suffix coefficient $A_{k+1,n}$ is positive, along that subsequence the surviving tied spine term has absolute value at least $(A_{k+1,n}/2)H_n(x)^m$. Thus persistent cancellation is already excluded by exact quarter-point facts and the binary-transition lemma; no additional Thue–Morse correlation conjecture remains.

Conjecture 14.3 (Direct inverse-spine asymptotic). *For $y = G_n(x)$ away from an explicit countable exceptional set, the derivatives of K_n admit an asymptotic expansion on the same universal quadratic scale Q_m ,*

$$K_n^{(m)}(y) = Q_m \left[\sum_{j=0}^{n-1} \tilde{A}_{j,n}(x) \tilde{B}_j(x)^m \Theta(2^m x_j) + o(\tilde{H}_n(x)^m) \right],$$

with computable inverse prefix weights $\tilde{B}_j$ obtained from the forward slopes and with a unique dominant term off finite tie surfaces.

The implicit identity $G_n \circ K_n = \text{id}$ and the same partition defect provide a plausible route, but one must control the recursive inverse Bell polynomials without assuming the desired envelope. A proof would upgrade abstract zero-radius transport to explicit derivative lower bounds for the inverse iterates. The finite-order replay does not establish the displayed scale, weights, exceptional set, or remainder.

Warning 14.4 (Former Conjecture 14.4 duplicates the forward report). The submitted statement that, for every fixed $z > 0$, the weighted partition sums

$$\sum_{\substack{\pi \in \Pi_m \\ 1 < |\pi| < m}} z^{s(\pi)} 2^{-D(\pi)}$$

admit a full asymptotic expansion whose leading term comes from the two first-defect shells: one block of size 2 with the rest singleton, and one block of size $m - 1$ with one singleton is exactly Conjecture 14.3 of the corrected forward-iterate report. It remains open manuscript mathematics there, with no exact Lean counterpart, but it is not a second independent inverse-iterate conjecture. A sharp version would give an explicit two-spine rate and the first correction term for both forward and prospective inverse spines.

Conjecture 14.5 (Nested Lambert phase law). *The repository's all-orders endpoint expansion for I can be iterated a fixed number of times to produce an all-orders transseries for K_n whose coefficient functions are smooth and periodic in a hierarchy of nested lower-branch Lambert phases. The leading term is (62), and each iteration transports rather than erases the dyadic periodic ripple.*

This all-orders claim is not implied by the proved leading scale, has not been tested by the interior Taylor-reversion experiment, and has no exact Lean counterpart.

14.1 Promising formalization route

A Lean formalization can be modular.

1. Define Q_m and formalize the partition defect and the exact factorization (26).
2. Prove a sufficiently explicit weighted defect bound for finite set partitions.
3. Use the existing derivative atlas and bounded-derivative API to formalize the two-spine reduction and the fixed- n orbit-spine induction.
4. Reuse strict monotonicity, convexity, and order-isomorphism modules to prove the finite tie-set and dense dominant-amplitude statements.
5. Finish the forward theorem with the existing power-series radius and analytic-at infrastructure.
6. Formalize the purely algebraic uniqueness of the compositional inverse of a formal power series with invertible linear coefficient.
7. Combine the analytic inverse-function theorem with formal-series convergence to prove [theorem 11.1](#) and transport E_n to Y_n .
8. Import the formal quadratic endpoint law and prove the elementary induction for [theorem 11.3](#).

The formal-reversion layer should be reusable well beyond the Fabius development.

14.2 Generalization to other atomic functions

The proof abstracts beyond the classical dyadic case. Suppose a smooth increasing map f has derivatives

$$f^{(m)}(x) = q^{\binom{m+1}{2}} \Phi(q^m x), \quad (69)$$

for some $q > 1$, with bounded Φ , recurrent nonvanishing along a dense set of q -adic orbits, and monotone orbit slopes yielding a unique dominant prefix. The same partition calculation replaces $2^{-D(\pi)}$ by $q^{-D(\pi)}$. Once forward iterates are controlled, [theorem 11.1](#) automatically transfers every zero-radius statement to inverse iterates. This suggests a general theorem for invertible Rvachev-type distribution functions and multiscale refinable maps.

15 Summary of the proof mechanism

The complete argument can be compressed into the following chain.

1. Exact differentiation gives $F^{(m)} = Q_m \Theta \circ (2^m \cdot)$ with $Q_m = 2^{m(m+1)/2}$ and $|\Theta| \leq 1$.
2. In Faà di Bruno's set-partition formula, every non-spine partition has a positive quadratic defect $D(\pi)$.
3. The weighted sum of all non-spine partitions is exponentially small despite Bell-number multiplicity.
4. Hence, after division by Q_m , the derivative $G_n^{(m)}$ is asymptotic to a finite sum of orbit spines:

$$A_{j,n} B_j^m \Theta(2^m x_j), \quad 0 \leq j < n.$$

5. Away from a finite tie set, monotone orbit dynamics makes one B_j uniquely maximal; away from countably many dyadic orbit preimages, binary transitions keep its amplitude at least $1/2$ infinitely often.
6. Therefore $R_{G_n}(x) = 0$ on a co-countable dense set E_n , and G_n is nowhere analytic.
7. Smooth local inverse Taylor series are unique formal compositional inverses. Positive convergence radius is invariant under reversion.
8. Thus $R_{K_n}(y) = 0$ on the co-countable dense set $Y_n = G_n(E_n)$, and no interior point of K_n can be analytic.
9. At the endpoints, $K_n(y)/y^\alpha \rightarrow \infty$ for every $\alpha > 0$; hence endpoint analyticity is impossible. At the center, the Taylor series is affine but does not represent the function.

This proves [theorems 1.1](#) and [1.2](#) and explains why finite self-composition cannot regularize either the Fabius function or its inverse.

A Auxiliary details on the partition defect

This appendix records two identities useful for sharpening [proposition 5.2](#).

Proposition A.1 (Exact minimum at fixed block count). *Among partitions of m with k blocks,*

$$\min D(\pi) = (k-1)(m-k). \quad (70)$$

Equality holds exactly when one block has size $m-k+1$ and every other block is a singleton.

Proof. For fixed m, k , formula (23) shows that minimizing D is equivalent to maximizing $\sum_i \binom{r_i}{2}$. Strict convexity of $r \mapsto \binom{r}{2}$ transfers one unit from a smaller nonsingleton block to a larger block and increases the sum. Iteration leaves one large block and $k-1$ singletons. Direct substitution gives (70). $\square$

Proposition A.2 (First defect shell). *For $m \geq 3$, the least positive defect is $m-2$. It occurs for exactly two block-size profiles:*

$$(2, 1, \dots, 1) \quad \text{and} \quad (m-1, 1). \quad (71)$$

Proof. From (70), the least positive value over $2 \leq k \leq m-1$ is

$$\min_{2 \leq k \leq m-1} (k-1)(m-k) = m-2,$$

attained only at $k=2$ and $k=m-1$. The equality classification in [proposition A.1](#) gives the two profiles. $\square$

This identifies the candidate leading correction to the spine sum. At the near-singleton end, choose one pair and leave all other elements singleton; at the near-giant end, choose the singleton outside a block of size $m-1$. Both carry the same dyadic penalty $2^{-(m-2)}$, but their derivative coefficients and orbit factors are different. A refined expansion should sum these shells before estimating the higher-defect remainder.

B Reproducibility guide

The archive contains the following files.

File	Purpose
<code>inverse_fabius_iterates_nowhere_analytic.tex</code>	Complete LaTeX source of this report.
<code>inverse_fabius_iterates_nowhere_analytic.pdf</code>	Rendered report.
<code>numerical_experiments.py</code>	Commented Python program for sinc-product reconstruction, forward Taylor composition, arbitrary-precision inverse-series reversion, spine evaluation, CSV output, and figures.
<code>numerical_output/spine_diagnostic.csv</code>	Raw forward derivative, spine, inverse coefficient, and formal-reversion residual diagnostics through the selected maximum order.
<code>numerical_output/numerical_metadata.txt</code>	Parameters, paired points, orbit points, slopes, prefixes, suffixes, dominant weight, and maximum formal-reversion residual.
<code>figures/*.png</code>	Figures included in the report.

A typical reproduction command is

```
python numerical_experiments.py \
  --output-dir numerical_output \
  --x0 0.437123456789 \
  --iterate-count 4 \
  --max-order 22
```

The program requires Python 3, NumPy, SciPy, Matplotlib, and mpmath. It performs consistency checks at $\text{up}(0)$, $\text{up}(1/2)$, $F(0)$, $F(1/2)$, and $F(1)$ before writing output.

C Repository source map

Under the common root `Analysis/FabiusFunction/`, the repository paths most directly connected to the proof are:

- `docs/Fabius_Function_and_Rvachev_Up/` for the primary exposition;
- `GlobalExtension.lean` and `BoundedDerivatives.lean` in `Lean/FabiusFunction/` for the derivative atlas;
- `Convexity.lean`, `Monotonicity.lean`, and the order-isomorphism modules for the orbit dynamics;
- `OriginalPaperSupplement.lean` and `NowhereAnalytic.lean` for the base nonanalyticity proof;
- `docs/Non_Elementarity_of_the_Fabius_Function/` for the exact analytic-locus obstruction for both F and I ;
- `docs/semi-formalized-research-frontiers/` for q-series, Bell/Bernoulli, sinc-product, Lambert- W , Legendre, and other frontier connections;
- `docs/archive/` and the frontier README for provenance and consolidation of historical TeX drafts.

References

- [1] J. Fabius, A probabilistic example of a nowhere analytic C^∞ -function, *Z. Wahrscheinlichkeitstheorie verw. Geb.* **5** (1966), 173–174.
- [2] J. Arias de Reyna, An infinitely differentiable function with compact support: definition and properties, English translation of *Rev. Real Acad. Ciencias Madrid* **76** (1982), 21–38, arXiv:1702.05442.
- [3] J. Arias de Reyna, Arithmetic of the Fabius function, arXiv:1702.06487v3, 2017.
- [4] L. Comtet, *Advanced Combinatorics: The Art of Finite and Infinite Expansions*, revised and enlarged edition, D. Reidel, 1974.
- [5] W. P. Johnson, The curious history of Faà di Bruno’s formula, *American Mathematical Monthly* **109** (2002), 217–234.
- [6] S. G. Krantz and H. R. Parks, *A Primer of Real Analytic Functions*, second edition, Birkhäuser, 2002.
- [7] P. Henrici, *Applied and Computational Complex Analysis, Vol. 1*, Wiley, 1974; see the chapters on power-series reversion and the analytic inverse-function theorem.
- [8] V. Reshetnikov, *The Fabius Function and Rvachev’s Up Function*, proof-backed primary exposition in the ProveIt repository, <https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction>, accessed 30 August 2026.